Relation-First Organization Theory: Resolution Structures, Proper Organization, and Persistence

Author: Artur Motruk
Affiliation: Independent researcher
ORCID: 0009-0003-6928-1701
Programme: Relation-First Organization Programme
Document role: Core foundation / resolution-structure layer
Public release status: Repository manuscript for public programme release

---

Foundation Statement

A relation-first organization structure is a quotient-presented resolution system. Its primitive carrier is a set of resolution occurrences. The answer face, question face, and zero-boundary face are quotient faces of that carrier. A total functional resolution map assigns each answer-question pair a zero-boundary value. Organization is the proper fixed-point layer generated by mutual derivation at a zero-boundary cut. Proper organization excludes both boundary extremes. Propagation is resolution preservation under covariant answer transport, contravariant question transport, and monotone zero-boundary transport. Persistence is the threshold-support of proper organizations across zero-boundary cuts. Evaluation is an ordering of proper organizations. Task-relative evaluation is obtained by deriving tasks as substructures of the primitive resolution carrier. The document connects persistence, propagation, task-relative evaluation, and evaluation by functoriality, stability, task-stability, evaluator-comparison theorems, barcode reconstruction, unlabeled bottleneck stability, interval-interleaving containment, and task/evaluator transport. It also states two self-limiting results: organization-internal data cannot recover structure-type distinctions not encoded in the resolution data, and quantitative stability on the zero-boundary cannot be defined from order alone.

---

0. Scope, Positioning, and Retention of the FCA Layer

0.1 Scope

This paper defines a mathematical foundation only.

No physical realization map is defined in this paper.

No claim is made here about spacetime, fields, particles, measurement, dynamics, probability, Hilbert spaces, Lagrangians, constants, or physical law.

0.2 Prior-Art Positioning

The cut construction in Sections 5–10 is a formal-context construction in the sense of Formal Concept Analysis.

The graded resolution map is a many-valued or lattice-valued context construction.

The fixed-value propagation law is a Chu-space morphism over a fixed value object.

The value-coarsening propagation law extends the fixed-value law by monotone maps between zero-boundary value faces.

The novelty claimed in this paper is not the Galois connection, the closure operators, or the concept lattice theorem. Those are standard consequences of the formal-context cut.

The commitments made here are:

1. quotient-face presentation from a primitive resolution carrier;
2. zero-boundary value face with forbidden extrema;

3. proper organization as the non-boundary fixed-point layer;
4. threshold-family and persistence analysis over zero-boundary cuts;
5. value-coarsening propagation with cut reflection conditions;
6. evaluator structure on proper organizations;
7. no-go and limitation results about admissibility and resolution-image complexity;
8. persistence functoriality under cut-isomorphism propagation;
9. stability bounds for persistence intervals under bounded perturbation;
10. evaluator-comparison and evaluator-nonuniqueness results;
11. barcode reconstruction and unlabeled bottleneck stability;
12. relation-internal task transport and evaluator-maximizer transport under cut-isomorphism propagation;
13. self-limitation of the face-level theory under direct-ternary representation equivalence;
14. structure-type invisibility no-go for distinctions not encoded in the resolution data;
15. metric-chain stability and metric-free stability limitation for the zero-boundary.

0.3 Why the FCA Layer Is Retained

The FCA-derived layer is retained explicitly, not replaced by the phrase "as in FCA," for four reasons.

First, the primitive data in this paper are not introduced as a pre-given formal context (G, M, I). The faces A_R, Q_R, Z_R are quotients of a primitive resolution carrier R. Therefore the formal context K_R^θ is derived at a cut, not primitive.

Second, the notation used later for boundary exclusion, value-coarsening, persistence, and evaluation depends on the exact cut-level derivation operators. Referring only to FCA would leave those dependencies implicit.

Third, the standard FCA theorems are not novelty claims. They are included as the translation layer that fixes what is inherited and what is added.

Fourth, the first-principles presentation makes the boundary and evaluator commitments inspectable. The document uses FCA machinery, but the intended object is not ordinary concept analysis; it is an organization theory built by adding boundary exclusion, threshold persistence, propagation, stability, and evaluator selection to the cut-context layer.

0.4 Strength of the Relation-First Claim

The relation-first claim in this paper is a quotient-face claim.

It does not eliminate the ambient identity of elements of the carrier set R.

It asserts only that the answer, question, and zero-boundary faces are not independently primitive; they are quotient faces of R.

A stronger carrier-free or internal-language formulation is not supplied in this paper.

0.5 Persistence Scope

Persistence is defined in this paper only when the zero-boundary face is a chain.

Persistence length is defined in two cases:

1. finite-chain length, by counting thresholds;
2. real-interval length, by Lebesgue measure.

For a general complete lattice which is not a chain, the cut family is still defined, but persistence length, barcode distance, and interval-interleaving containment are defined here only after additional chain or real-interval structure is supplied.

0.6 Task Relativity

This paper formalizes task relativity only in relation-internal form.

A task is not an additional primitive.

A task is a finite positive-negative substructure of the primitive resolution carrier R.

Task-relative evaluators are evaluators restricted to proper organizations satisfying such relation-internal task constraints.

No external category of tasks is introduced in this paper.

---

1. Logical and Order-Theoretic Conventions

A set is a collection in the ambient metatheory.

For a set X,

$$\mathcal{P}(X) := \{S : S \subseteq X\}.$$

For a relation $\sim$ on a set X,

$$\sim \text{ is an equivalence relation}$$

means

$$\forall x \in X,\ x \sim x,$$

$$\forall x, y \in X, \quad x \sim y \Rightarrow y \sim x,$$

and

$$\forall x, y, z \in X, \quad ((x \sim y) \wedge (y \sim z)) \Rightarrow x \sim z.$$

For an equivalence relation $\sim$ on X,

$$X/{\sim} := \{[x]_\sim : x \in X\},$$

where

$$[x]_\sim := \{y \in X : y \sim x\}.$$

For a partially ordered set $(L, \leq)$,

$$\bigwedge S$$

means the greatest lower bound of $S \subseteq L$, when it exists, and

$$\bigvee S$$

means the least upper bound of $S \subseteq L$, when it exists.

A complete lattice is a partially ordered set $(L, \leq)$ such that every subset $S \subseteq L$ has $\bigwedge S$ and $\bigvee S$.

A chain is a partially ordered set $(L, \leq)$ such that

$$\forall x, y \in L, \quad x \leq y \text{ or } y \leq x.$$

For $x, y \in L$,

$$x < y$$

means

$$(x \leq y) \wedge (x \neq y).$$

For a finite set X,

$$|X|$$

means the cardinality of X.

For a map $h : X \to Y$ and a subset $S \subseteq X$,

$$h[S] := \{h(x) : x \in S\}.$$

For a map $h : X \to Y$ and a subset $T \subseteq Y$,

$$h^{-1}[T] := \{x \in X : h(x) \in T\}.$$

---

2. Primitive Resolution Carrier

Definition 2.1: Organization Carrier

An **organization carrier** is a quintuple

$$\mathbb{R} := (R, \equiv_A, \equiv_Q, \equiv_Z, \preceq_R)$$

such that

$$R \text{ is a nonempty set,}$$

$$\equiv_A, \equiv_Q, \equiv_Z$$

are equivalence relations on R, and

$$\preceq_R \subseteq (R/{\equiv_Z}) \times (R/{\equiv_Z}).$$

Definition 2.2: Answer Face

$$A_R := R/{\equiv_A}.$$

For $r \in R$,

$$\alpha_R(r) := [r]_{\equiv_A} \in A_R.$$

Definition 2.3: Question Face

$$Q_R := R/{\equiv_Q}.$$

For $r \in R$,

$$\chi_R(r) := [r]_{\equiv_Q} \in Q_R.$$

Definition 2.4: Zero-Boundary Face

$$Z_R := R/{\equiv_Z}.$$

For $r \in R$,

$$\zeta_R(r) := [r]_{\equiv_Z} \in Z_R.$$

Definition 2.5: Organization Structure

An **organization structure** is an organization carrier $\mathbb{R}$ satisfying Axioms R1, R2, Z1, Z2, and Z3.

---

3. Resolution Axioms, Resolution Map, and Ternary Representation

Axiom R1: Total Pair Coverage

$$\forall a \in A_R \ \forall q \in Q_R \ \exists r \in R, \quad \alpha_R(r) = a \wedge \chi_R(r) = q.$$

Axiom R2: Functional Zero-Boundary Resolution

$$\forall r, s \in R, \quad (\alpha_R(r) = \alpha_R(s) \wedge \chi_R(r) = \chi_R(s)) \Rightarrow \zeta_R(r) = \zeta_R(s).$$

Definition 3.1: Resolution Map

$$\rho_R : A_R \times Q_R \to Z_R$$

is defined by

$$\rho_R(a, q) := \zeta_R(r),$$

where $r \in R$ satisfies

$$\alpha_R(r) = a, \quad \chi_R(r) = q.$$

Theorem 3.2: Well-Defined Resolution Map

$$\rho_R$$

is a total function from $A_R \times Q_R$ to Z_R.

Proof

Let

$$a \in A_R, \quad q \in Q_R.$$

By Axiom R1, there exists $r \in R$ with

$$\alpha_R(r) = a, \quad \chi_R(r) = q.$$

Thus $\rho_R(a, q)$ has at least one assigned value.

Let $r, s \in R$ both satisfy

$$\alpha_R(r) = a, \quad \chi_R(r) = q,$$

and

$$\alpha_R(s) = a, \quad \chi_R(s) = q.$$

Then

$$\alpha_R(r) = \alpha_R(s), \quad \chi_R(r) = \chi_R(s).$$

By Axiom R2,

$$\zeta_R(r) = \zeta_R(s).$$

Therefore the value assigned to $\rho_R(a, q)$ is independent of the witness r.

Hence ρ_R is total and well-defined.

$$\square$$

Definition 3.3: Direct Ternary Presentation

A **direct ternary presentation** is a quintuple

$$(A, Q, Z, \preceq, \rho)$$

such that A, Q, Z are nonempty sets, $\preceq$ is an order on Z, and

$$\rho : A \times Q \to Z$$

is a total function.

Definition 3.4: Induced Direct Ternary Presentation

The direct ternary presentation induced by an organization structure R is

$$D(R) := (A_R, Q_R, Z_R, \preceq_R, \rho_R).$$

Definition 3.5: Quotient Realization of a Direct Ternary Presentation

Given a direct ternary presentation

$$D = (A, Q, Z, \preceq, \rho),$$

define

$$R_D := A \times Q.$$

For $(a, q), (a', q') \in R_D$, define

$$(a, q) \equiv_A (a', q') \quad \Longleftrightarrow \quad a = a',$$

$$(a, q) \equiv_Q (a', q') \quad \Longleftrightarrow \quad q = q',$$

and

$$(a, q) \equiv_Z (a', q') \quad \Longleftrightarrow \quad \rho(a, q) = \rho(a', q').$$

Define $\preceq_{R_D}$ on $R_D/{\equiv_Z}$ by

$$[(a, q)]_{\equiv_Z} \preceq_{R_D} [(a', q')]_{\equiv_Z} \quad \Longleftrightarrow \quad \rho(a, q) \preceq \rho(a', q').$$

Theorem 3.6: Representation Equivalence at the Level of Faces

For every direct ternary presentation $D = (A, Q, Z, \preceq, \rho)$, the quotient realization R_D induces faces canonically isomorphic to $A, Q, \mathrm{Im}(\rho)$, and its resolution map agrees with ρ after these identifications.

Proof

The map

$$A_{R_D} \to A, \quad [(a, q)]_{\equiv_A} \mapsto a$$

is well-defined and bijective by the definition of $\equiv_A$.

The map

$$Q_{R_D} \to Q, \quad [(a, q)]_{\equiv_Q} \mapsto q$$

is well-defined and bijective by the definition of $\equiv_Q$.

The map

$$Z_{R_D} \to \mathrm{Im}(\rho), \quad [(a,q)]_{\equiv_Z} \mapsto \rho(a,q)$$

is well-defined and bijective by the definition of $\equiv_Z$.

Under these identifications,

$$\rho_{R_D}([(a,q)]_{\equiv_A}, [(a,q)]_{\equiv_Q}) = [(a,q)]_{\equiv_Z},$$

which corresponds to $\rho(a,q)$.

$\square$

Remark 3.7: Consequence for the Relation-First Claim

All cut-level, closure-level, persistence-level, and evaluator-level theorems in this paper depend on $D(R)$.

The quotient presentation is therefore a foundational presentation choice, not a theorem-strengthening device in this paper.

Definition 3.8: Face-Level Construction

A construction $\mathcal{C}$ on organization structures is face-level when, for every pair of organization structures R, R', every isomorphism of direct ternary presentations

$$\iota_A : A_R \to A_{R'}, \qquad \iota_Q : Q_R \to Q_{R'}, \qquad \iota_Z : Z_R \to Z_{R'}$$

with

$$\iota_Z(\rho_R(a,q)) = \rho_{R'}(\iota_A(a), \iota_Q(q))$$

and

$$z \preceq_R z' \quad \Longleftrightarrow \quad \iota_Z(z) \preceq_{R'} \iota_Z(z')$$

induces an isomorphism

$$\mathcal{C}(R) \cong \mathcal{C}(R').$$

Theorem 3.9: Face-Level Invariance No-Go

No face-level construction can distinguish two quotient-presented organization structures with isomorphic direct ternary presentations.

Consequently, no theorem using only

$$A_R, Q_R, Z_R, \preceq_R, \rho_R$$

can establish a stronger relation-first result than its direct-ternary reformulation.

Proof

Let $\mathcal{C}$ be face-level.

Let R, R' have isomorphic direct ternary presentations.

By Definition 3.8, the isomorphism of direct ternary presentations induces

$$\mathcal{C}(R) \cong \mathcal{C}(R').$$

Thus $\mathcal{C}$ cannot distinguish R from R'.

Every theorem using only

$$A_R, Q_R, Z_R, \preceq_R, \rho_R$$

is invariant under this isomorphism.

Therefore such a theorem has an equivalent direct-ternary reformulation.

□

Corollary 3.10: Self-Limitation of the Present Foundation

The present paper does not prove that the quotient carrier is mathematically stronger than a direct ternary presentation.

It proves only face-level organization theorems.

A theorem that essentially uses carrier-level data must use structure not determined by

$$A_R, Q_R, Z_R, \preceq_R, \rho_R.$$

Proof

This is Theorem 3.9 applied to all face-level constructions used below.

□

---

4. Zero-Boundary Order

4.1 Terminological Definition

The term **zero-boundary** is a defined name for the ordered value face $(Z_R, \preceq_R)$.

It means:

$$\text{the resolution-value face with a least boundary } z_R^- \text{ and a greatest boundary } z_R^+.$$

It does not mean topological boundary, homological boundary, temporal boundary, manifold boundary, or physical boundary.

The word **zero** marks the lower resolution boundary z_R^-.

The word **boundary** marks the fact that both extremal values z_R^- and z_R^+ are excluded from proper organization.

Axiom Z1: Complete Zero-Boundary Lattice

$$(Z_R, \preceq_R)$$

is a complete lattice.

Definition 4.2: Lower Boundary

$$z_R^- := \bigwedge Z_R.$$

Definition 4.3: Upper Boundary

$$z_R^+ := \bigvee Z_R.$$

Axiom Z2: Nondegenerate Zero-Boundary

$$z_R^- \neq z_R^+.$$

Definition 4.4: Strict Boundary Order

For $x, y \in Z_R$,

$$x \prec_R y \quad \Longleftrightarrow \quad (x \preceq_R y) \wedge (x \neq y).$$

Definition 4.5: Proper Zero-Boundary Interior

$$Z_R^\circ := \{z \in Z_R : z_R^- \prec_R z \prec_R z_R^+\}.$$

Axiom Z3: Nonempty Proper Interior

$$Z_R^\circ \neq \emptyset.$$

---

5. Cut Contexts

Let

$$\theta \in Z_R.$$

Definition 5.1: θ-Resolution Incidence

For $a \in A_R$ and $q \in Q_R$,

$$a \Vdash_R^\theta q \quad \Longleftrightarrow \quad \theta \preceq_R \rho_R(a, q).$$

Definition 5.2: θ-Cut Context

$$I_R^\theta := \{(a, q) \in A_R \times Q_R : a \Vdash_R^\theta q\}.$$

Definition 5.3: Induced Formal Context

$$K_R^\theta := (A_R, Q_R, I_R^\theta).$$

---

6. Mutual Derivation Operators

Let

$$S \subseteq A_R, \quad T \subseteq Q_R.$$

Definition 6.1: Up-Derivation

$$S^{\uparrow_R^\theta} := \{q \in Q_R : \forall a \in S,\ a \Vdash_R^\theta q\}.$$

Definition 6.2: Down-Derivation

$$T^{\downarrow_R^\theta} := \{a \in A_R : \forall q \in T,\ a \Vdash_R^\theta q\}.$$

---

7. FCA Translation Theorems

Theorem 7.1: Mutual Derivation Law

For all

$$S \subseteq A_R, \quad T \subseteq Q_R, \quad \theta \in Z_R,$$

$$S \subseteq T^{\downarrow_R^\theta} \quad \Longleftrightarrow \quad T \subseteq S^{\uparrow_R^\theta}.$$

Proof

$$S \subseteq T^{\downarrow_R^\theta}$$

$$\Longleftrightarrow \forall a \in S,\ a \in T^{\downarrow_R^\theta}$$

$$\Longleftrightarrow \forall a \in S\ \forall q \in T,\ a \Vdash_R^\theta q$$

$$\Longleftrightarrow \forall q \in T\ \forall a \in S,\ a \Vdash_R^\theta q$$

$$\Longleftrightarrow \forall q \in T,\ q \in S^{\uparrow_R^\theta}$$

$$\Longleftrightarrow T \subseteq S^{\uparrow_R^\theta}.$$

$$\square$$

Theorem 7.2: Up-Antitonicity

For all

$$S, S' \subseteq A_R, \quad \theta \in Z_R,$$

$$S \subseteq S' \quad \Longrightarrow \quad (S')^{\uparrow_R^\theta} \subseteq S^{\uparrow_R^\theta}.$$

Proof

Let

$$q \in (S')^{\uparrow_R^\theta}.$$

Then

$$\forall a \in S',\ a \Vdash_R^\theta q.$$

If $S \subseteq S'$, then

$$\forall a \in S,\ a \Vdash_R^\theta q.$$

Therefore

$$q \in S^{\uparrow_R^\theta}.$$

$$\square$$

Theorem 7.3: Down-Antitonicity

For all

$$T, T' \subseteq Q_R, \quad \theta \in Z_R,$$

$$T \subseteq T' \quad \Longrightarrow \quad (T')^{\downarrow_R^\theta} \subseteq T^{\downarrow_R^\theta}.$$

Proof

Let

$$a \in (T')^{\downarrow_R^\theta}.$$

Then

$$\forall q \in T',\ a \Vdash_R^\theta q.$$

If $T \subseteq T'$, then

$$\forall q \in T,\ a \Vdash_R^\theta q.$$

Therefore

$$a \in T^{\downarrow_R^\theta}.$$

$$\square$$

---

8. Compression Operators

Definition 8.1: Answer-Compression Operator

$$C^{\theta}_{A,R} : \mathcal{P}(A_R) \to \mathcal{P}(A_R)$$

is defined by

$$C^{\theta}_{A,R}(S) := S^{\uparrow^{\theta}_{R}\downarrow^{\theta}_{R}}.$$

Definition 8.2: Question-Compression Operator

$$C^{\theta}_{Q,R} : \mathcal{P}(Q_R) \to \mathcal{P}(Q_R)$$

is defined by

$$C^{\theta}_{Q,R}(T) := T^{\downarrow^{\theta}_{R}\uparrow^{\theta}_{R}}.$$

Theorem 8.3: Compression Operators are Closure Operators

$$C^{\theta}_{A,R}$$

and

$$C^{\theta}_{Q,R}$$

are extensive, monotone, and idempotent.

Proof

This is the standard closure theorem for a Galois connection.

For extensivity of $C^{\theta}_{A,R}$, let $T = S^{\uparrow^{\theta}_{R}}$. Since

$$T \subseteq S^{\uparrow^{\theta}_{R}},$$

Theorem 7.1 gives

$$S \subseteq T^{\downarrow^{\theta}_{R}} = C^{\theta}_{A,R}(S).$$

For extensivity of $C^{\theta}_{Q,R}$, let $S = T^{\downarrow^{\theta}_{R}}$. Since

$$S \subseteq T^{\downarrow^{\theta}_{R}},$$

Theorem 7.1 gives

$$T \subseteq S^{\uparrow^{\theta}_{R}} = C^{\theta}_{Q,R}(T).$$

Monotonicity follows by applying Theorems 7.2 and 7.3 in succession.

Idempotence follows from extensivity, monotonicity, and antitonicity of the derivation operators: if $D = S^{\uparrow^{\theta}_{R}\downarrow^{\theta}_{R}}$, then $S \subseteq D$, so

$$D^{\uparrow^\theta_R} \subseteq S^{\uparrow^\theta_R}.$$

Also $D = S^{\uparrow^\theta_R \downarrow^\theta_R}$ and Theorem 7.1 give

$$S^{\uparrow^\theta_R} \subseteq D^{\uparrow^\theta_R}.$$

Thus

$$D^{\uparrow^\theta_R} = S^{\uparrow^\theta_R},$$

so

$$C^\theta_{A,R}(D) = D.$$

The proof for $C^\theta_{Q,R}$ is dual.

$\square$

---

9. Organizations and Organization Lattice

Definition 9.1: θ-Organization

$$\mathrm{Org}_\theta(R) := \{(S,T) \in \mathcal{P}(A_R) \times \mathcal{P}(Q_R) : S = T^{\downarrow^\theta_R} \wedge T = S^{\uparrow^\theta_R}\}.$$

Definition 9.2: Organization Order

For

$$(S,T), (S',T') \in \mathrm{Org}_\theta(R),$$

define

$$(S,T) \leq_\theta (S',T') \quad \Longleftrightarrow \quad S \subseteq S'.$$

Theorem 9.3: Dual Question Order

For

$$(S,T), (S',T') \in \mathrm{Org}_\theta(R),$$

$$(S,T) \leq_\theta (S',T') \quad \Longleftrightarrow \quad T' \subseteq T.$$

Proof

If $S \subseteq S'$, then Theorem 7.2 gives

$$(S')^{\uparrow_R^\theta} \subseteq S^{\uparrow_R^\theta}.$$

Since $T' = (S')^{\uparrow_R^\theta}$ and $T = S^{\uparrow_R^\theta}$,

$$T' \subseteq T.$$

The converse follows dually from Theorem 7.3.

$$\square$$

Theorem 9.4: Complete Organization Lattice

$$(\mathrm{Org}_\theta(R), \leq_\theta)$$

is a complete lattice.

For any family

$$\{(S_i, T_i) : i \in I\} \subseteq \mathrm{Org}_\theta(R),$$

$$\bigwedge_{i\in I}(S_i, T_i) = \left(\bigcap_{i\in I} S_i, \left(\bigcap_{i\in I} S_i \right)^{\uparrow_R^\theta} \right),$$

and

$$\bigvee_{i\in I}(S_i, T_i) = \left(\left(\bigcup_{i\in I} S_i \right)^{\uparrow_R^\theta \downarrow_R^\theta}, \left(\bigcup_{i\in I} S_i \right)^{\uparrow_R^\theta} \right).$$

Proof

This is the standard concept-lattice theorem for the formal context K_R^θ.

For the meet formula, set

$$M := \bigcap_{i\in I} S_i, \quad N := M^{\uparrow_R^\theta}.$$

The Galois law gives

$$N^{\downarrow_R^\theta} = M.$$

Thus $(M, N) \in \mathrm{Org}_\theta(R)$. It is the greatest lower bound by inclusion of answer components.

For the join formula, set

$$U := \bigcup_{i\in I} S_i, \quad J := U^{\uparrow_R^\theta \downarrow_R^\theta}, \quad K := U^{\uparrow_R^\theta}.$$

The closure theorem gives $(J, K) \in \mathrm{Org}_\theta(R)$. It is the least upper bound by inclusion of answer components.

$\square$

---

10. Boundary Organizations and Proper Organization

Definition 10.1: Lower Boundary Organization

$$\bot_\theta(R) := (Q_R^{\downarrow_R^\theta}, Q_R).$$

Definition 10.2: Upper Boundary Organization

$$\top_\theta(R) := (A_R, A_R^{\uparrow_R^\theta}).$$

Theorem 10.3: Boundary Elements

$$\bot_\theta(R) = \bigwedge \mathrm{Org}_\theta(R),$$

and

$$\top_\theta(R) = \bigvee \mathrm{Org}_\theta(R).$$

Proof

Let

$$(S, T) \in \mathrm{Org}_\theta(R).$$

Since $T \subseteq Q_R$, Theorem 7.3 gives

$$Q_R^{\downarrow_R^\theta} \subseteq T^{\downarrow_R^\theta} = S.$$

Thus

$$\bot_\theta(R) \leq_\theta (S, T).$$

Since $S \subseteq A_R$,

$$(S, T) \leq_\theta \top_\theta(R).$$

Thus $\bot_\theta(R)$ is least and $\top_\theta(R)$ is greatest.

$\square$

Definition 10.4: Proper Organization Layer

$$\mathrm{Org}_\theta^\circ(R) := \mathrm{Org}_\theta(R) \setminus \{\bot_\theta(R), \top_\theta(R)\}.$$

Definition 10.5: Admissible Organization Structure

$$R \text{ is admissible}$$

means

$$\exists\theta \in Z_R^\circ, \quad \mathrm{Org}_\theta^\circ(R) \neq \emptyset.$$

Remark 10.6: Boundary Excision

Boundary excision is not used to prove the FCA translation theorems.

Boundary excision is used only in definitions involving proper organization, admissibility, persistence support, and evaluators on proper organizations.

Theorems 11.3 and 11.4 show that the universal and empty cuts produce no proper organization under this definition.

Remark 10.7: Defense of Boundary Excision

Boundary excision is the formal expression of a nondegeneracy requirement.

At the lower boundary, no answer-question pair resolves at the chosen cut. The derivation operators then collapse to the same boundary behavior for every internal pattern of ρ_R.

At the upper boundary, every answer-question pair resolves at the chosen cut. The derivation operators again cease to detect the internal pattern of ρ_R.

Thus both extrema are organizationally insensitive. They may exist as boundary objects of the full concept lattice, but they do not witness organization as internal contrast.

The proper layer is therefore not introduced to deny the boundary elements. It is introduced to separate pattern-sensitive fixed points from the two degenerate fixed points determined by total absence or total saturation of incidence.

---

11. No-Go and Limitation Results

Definition 11.1: Universal Cut

$$I_R^\theta = A_R \times Q_R.$$

Definition 11.2: Empty Cut

$$I_R^\theta = \emptyset.$$

Theorem 11.3: Universal Cut Has No Proper Organization

If

$$I_R^\theta = A_R \times Q_R,$$

then

$$\mathrm{Org}_\theta^\circ(R) = \emptyset.$$

Proof

If $I_R^\theta = A_R \times Q_R$, then for all $S \subseteq A_R$,

$$S^{\uparrow_R^\theta} = Q_R,$$

and for all $T \subseteq Q_R$,

$$T^{\downarrow_R^\theta} = A_R.$$

Therefore

$$\mathrm{Org}_\theta(R) = \{(A_R, Q_R)\}.$$

The only organization is boundary.

$$\square$$

Theorem 11.4: Empty Cut Has No Proper Organization

If

$$I_R^\theta = \emptyset,$$

then

$$\mathrm{Org}_\theta^\circ(R) = \emptyset.$$

Proof

If $I_R^\theta = \emptyset$, then the only organizations are

$$(\emptyset, Q_R)$$

and

$$(A_R, \emptyset).$$

These are $\bot_\theta(R)$ and $\top_\theta(R)$.

Therefore

$$\mathrm{Org}_\theta^\circ(R) = \emptyset.$$

$$\square$$

Corollary 11.5: Constant Resolution No-Go

If there exists $z_0 \in Z_R$ such that

$$\forall a \in A_R \ \forall q \in Q_R, \quad \rho_R(a, q) = z_0,$$

then for every $\theta \in Z_R$,

$$\mathrm{Org}_\theta^\circ(R) = \emptyset.$$

Proof

Let $\theta \in Z_R$.

If

$$\theta \preceq_R z_0,$$

then

$$I_R^\theta = A_R \times Q_R,$$

so Theorem 11.3 applies.

If

$$\neg(\theta \preceq_R z_0),$$

then

$$I_R^\theta = \emptyset,$$

so Theorem 11.4 applies.

$\square$

Definition 11.6: Resolution Image

$$\mathrm{Im}(\rho_R) := \{\rho_R(a, q) : a \in A_R, q \in Q_R\}.$$

Theorem 11.7: Minimum Image-Cardinality for Admissibility

If R is admissible, then

$$|\,\mathrm{Im}(\rho_R)| \geq 2.$$

Proof

If

$$|\operatorname{Im}(\rho_R)| = 1,$$

then ρ_R is constant.

By Corollary 11.5,

$$\operatorname{Org}^{\circ}_{\theta}(R) = \emptyset$$

for every $\theta \in Z_R$.

Thus R is not admissible.

The contrapositive gives the result.

$\square$

Theorem 11.8: No Image-Cardinality-Only Upper Bound on Proper Organizations

There is no function

$$f : \mathbb{N} \to \mathbb{N}$$

such that every finite organization structure R satisfies

$$|\operatorname{Org}^{\circ}_{\theta}(R)| \leq f(|\operatorname{Im}(\rho_R)|)$$

for all $\theta \in Z_R$.

Proof

Let $n \geq 2$.

Define

$$A_R := \{a_1, \dots, a_n\}, \quad Q_R := \{q_1, \dots, q_n\},$$

and

$$Z_R := \{z_-, z_0, z_+\}, \quad z_- \prec_R z_0 \prec_R z_+.$$

Let $\theta := z_0$, and define

$$\rho_R(a_i, q_j) = z_0 \quad \Longleftrightarrow \quad i = j,$$

and

$$\rho_R(a_i, q_j) = z_- \quad \Longleftrightarrow \quad i \neq j.$$

Then

$$|\operatorname{Im}(\rho_R)| = 2.$$

At $\theta = z_0$, the incidence relation is equality:

$$a_i \Vdash_R^{z_0} q_j \quad \Longleftrightarrow \quad i = j.$$

The proper organizations include

$$(\{a_i\}, \{q_i\})$$

for each $1 \le i \le n$.

Therefore

$$|\operatorname{Org}^{\circ}_{z_0}(R)| \ge n.$$

Since n is arbitrary and $|\operatorname{Im}(\rho_R)| = 2$ is fixed, no such function f exists.

$\square$

Corollary 11.9: Correct Form of the Low-Image Question

Image cardinality alone cannot bound organization richness.

The valid low-image statement supplied by this framework is Theorem 11.7.

Additional richness bounds require additional constraints beyond $|\operatorname{Im}(\rho_R)|$, such as restrictions on incidence-pattern rank, cut-family homology, lattice shape, or allowed equivalence-relation structure on R.

Remark 11.10: Pattern Over Vocabulary

Theorem 11.8 shows that organization richness is not controlled by the number of available resolution values alone.

The controlling object is the pattern by which resolution values are distributed over $A_R \times Q_R$.

---

12. Value-Coarsening Propagation

Definition 12.1: Monotone Zero-Boundary Map

For organization structures R, R', a map

$$k : Z_R \to Z_{R'}$$

is monotone when

$$\forall x, y \in Z_R, \quad x \preceq_R y \Rightarrow k(x) \preceq_{R'} k(y).$$

Definition 12.2: Order-Reflecting Zero-Boundary Map

A map

$$k : Z_R \to Z_{R'}$$

is order-reflecting when

$$\forall x, y \in Z_R, \quad k(x) \preceq_{R'} k(y) \Rightarrow x \preceq_R y.$$

Definition 12.3: Order-Embedding Zero-Boundary Map

A map

$$k : Z_R \to Z_{R'}$$

is an order embedding when it is monotone and order-reflecting.

Definition 12.4: Interior-Preserving Zero-Boundary Map

A map

$$k : Z_R \to Z_{R'}$$

is interior-preserving when

$$k[Z_R^\circ] \subseteq Z_{R'}^\circ.$$

Definition 12.5: Value-Coarsening Propagation Morphism

A value-coarsening propagation morphism

$$(f, g, k) : R \to R'$$

is a triple of maps

$$f : A_R \to A_{R'},$$

$$g : Q_{R'} \to Q_R,$$

$$k : Z_R \to Z_{R'}$$

such that k is monotone and

$$\forall a \in A_R \ \forall q' \in Q_{R'}, \quad k(\rho_R(a, g(q'))) = \rho_{R'}(f(a), q').$$

Theorem 12.6: Forward Cut Preservation

Let

$$(f, g, k) : R \to R'$$

be a value-coarsening propagation morphism.

For every $\theta \in Z_R$,

$$a \Vdash_R^{\theta} g(q') \quad \Longrightarrow \quad f(a) \Vdash_{R'}^{k(\theta)} q'.$$

Proof

Assume

$$a \Vdash_R^{\theta} g(q').$$

Then

$$\theta \preceq_R \rho_R(a, g(q')).$$

Since k is monotone,

$$k(\theta) \preceq_{R'} k(\rho_R(a, g(q'))).$$

By Definition 12.5,

$$k(\rho_R(a, g(q'))) = \rho_{R'}(f(a), q').$$

Therefore

$$k(\theta) \preceq_{R'} \rho_{R'}(f(a), q'),$$

so

$$f(a) \Vdash_{R'}^{k(\theta)} q'.$$

$$\square$$

Theorem 12.7: Reflected Cut Preservation

Let

$$(f, g, k) : R \to R'$$

be a value-coarsening propagation morphism.

If k is order-reflecting, then for every $\theta \in Z_R$,

$$f(a) \Vdash_{R'}^{k(\theta)} q' \quad \Longrightarrow \quad a \Vdash_R^{\theta} g(q').$$

Proof

Assume

$$f(a) \Vdash_{R'}^{k(\theta)} q'.$$

Then

$$k(\theta) \preceq_{R'} \rho_{R'}(f(a), q').$$

By Definition 12.5,

$$\rho_{R'}(f(a), q') = k(\rho_R(a, g(q'))).$$

Therefore

$$k(\theta) \preceq_{R'} k(\rho_R(a, g(q'))).$$

Since k is order-reflecting,

$$\theta \preceq_R \rho_R(a, g(q')).$$

Thus

$$a \Vdash_R^{\theta} g(q').$$

□

Corollary 12.8: Cut Equivalence Under Order Embedding

If k is an order embedding, then

$$a \Vdash_R^{\theta} g(q') \quad \Longleftrightarrow \quad f(a) \Vdash_{R'}^{k(\theta)} q'.$$

Proof

The forward implication is Theorem 12.6.

The reverse implication is Theorem 12.7.

□

Definition 12.9: Fixed-Value Propagation Morphism

A fixed-value propagation morphism is a value-coarsening propagation morphism

$$(f, g, k) : R \to R'$$

such that

$$Z_R = Z_{R'}, \quad \preceq_R = \preceq_{R'}, \quad k = \mathrm{id}_{Z_R}.$$

Corollary 12.10: Fixed-Value Cut Equivalence

If $(f, g, k) : R \to R'$ is fixed-value, then for every $\theta \in Z_R$,

$$a \Vdash^{\theta}_{R} g(q') \quad \Longleftrightarrow \quad f(a) \Vdash^{\theta}_{R'} q'.$$

Proof

For fixed-value propagation,

$$k = \mathrm{id}_{Z_R}.$$

The identity map is an order embedding.

Apply Corollary 12.8.

$\square$

Definition 12.11: Identity Propagation Morphism

$$\mathrm{id}_R := (\mathrm{id}_{A_R}, \mathrm{id}_{Q_R}, \mathrm{id}_{Z_R}).$$

Definition 12.12: Composition of Propagation Morphisms

For

$$(f, g, k) : R \to R',$$

and

$$(f', g', k') : R' \to R'',$$

define

$$(f', g', k') \circ (f, g, k) := (f' \circ f,\ g \circ g',\ k' \circ k).$$

Theorem 12.13: Propagation Category

Organization structures and value-coarsening propagation morphisms form a category.

Proof

Identity follows from identity laws for functions and

$$\mathrm{id}_{Z_R}(\rho_R(a, q)) = \rho_R(a, q).$$

Composition follows from

$$(k' \circ k)(\rho_R(a, (g \circ g')(q''))) = \rho_{R''}((f' \circ f)(a), q'')$$

by two applications of Definition 12.5.

Associativity and unit laws follow from associativity and unit laws for function composition.

$\square$

Theorem 12.14: Order-Embedding Morphisms Form a Subcategory

The value-coarsening propagation morphisms whose zero-boundary maps are order embeddings form a subcategory.

Proof

The identity map on any ordered set is an order embedding.

Let

$$k : Z_R \to Z_{R'}, \quad k' : Z_{R'} \to Z_{R''}$$

be order embeddings.

If

$$x \preceq_R y,$$

then

$$k(x) \preceq_{R'} k(y),$$

and therefore

$$k'(k(x)) \preceq_{R''} k'(k(y)).$$

Thus $k' \circ k$ is monotone.

If

$$(k' \circ k)(x) \preceq_{R''} (k' \circ k)(y),$$

then order-reflection of k' gives

$$k(x) \preceq_{R'} k(y),$$

and order-reflection of k gives

$$x \preceq_R y.$$

Thus $k' \circ k$ is order-reflecting.

Therefore the class is closed under identity and composition.

$\square$

Definition 12.15: Cut-Isomorphism Propagation Morphism

A value-coarsening propagation morphism

$$m = (f, g, k) : R \to R'$$

is a cut-isomorphism propagation morphism when:

1. $f : A_R \to A_{R'}$ is bijective;
2. $g : Q_{R'} \to Q_R$ is bijective;
3. $k : Z_R \to Z_{R'}$ is an interior-preserving order embedding.

Definition 12.16: Organization Transport Along a Cut-Isomorphism

For a cut-isomorphism propagation morphism

$$m = (f, g, k) : R \to R'$$

and

$$O = (S, T) \in \mathcal{P}(A_R) \times \mathcal{P}(Q_R),$$

define

$$m_*(O) := (f[S], g^{-1}[T]).$$

---

13. Threshold Families and Persistence

Assume in this section that $(Z_R, \preceq_R)$ is a chain unless stated otherwise.

Theorem 13.1: Cut Antitonicity

For

$$\theta, \phi \in Z_R, \quad \theta \preceq_R \phi,$$

$$I_R^\phi \subseteq I_R^\theta.$$

Proof

Let

$$(a, q) \in I_R^\phi.$$

Then

$$\phi \preceq_R \rho_R(a, q).$$

Since

$$\theta \preceq_R \phi,$$

transitivity gives

$$\theta \preceq_R \rho_R(a,q).$$

Thus

$$(a,q) \in I_R^\theta.$$

$$\square$$

Theorem 13.2: Finite Stepwise Constancy

Assume

$$A_R \times Q_R$$

is finite.

Let

$$V_R := \mathrm{Im}(\rho_R).$$

If $\theta, \phi \in Z_R$ satisfy

$$\forall v \in V_R, \quad (\theta \preceq_R v \Longleftrightarrow \phi \preceq_R v),$$

then

$$I_R^\theta = I_R^\phi,$$

and

$$\mathrm{Org}_\theta(R) = \mathrm{Org}_\phi(R).$$

Proof

For every $(a,q) \in A_R \times Q_R$,

$$\rho_R(a,q) \in V_R.$$

Thus

$$\theta \preceq_R \rho_R(a,q) \quad \Longleftrightarrow \quad \phi \preceq_R \rho_R(a,q).$$

Hence

$$(a,q) \in I_R^\theta \quad \Longleftrightarrow \quad (a,q) \in I_R^\phi.$$

Therefore

$$I_R^\theta = I_R^\phi.$$

All derivation operators and organizations are determined by incidence.

$\square$

Corollary 13.3: Threshold-Complexity Bound

If

$$A_R \times Q_R$$

is finite and $(Z_R, \preceq_R)$ is a chain, then the family

$$\{I_R^\theta : \theta \in Z_R\}$$

has cardinality at most

$$|\operatorname{Im}(\rho_R)| + 1.$$

The family

$$\{\operatorname{Org}_\theta(R) : \theta \in Z_R\}$$

also has cardinality at most

$$|\operatorname{Im}(\rho_R)| + 1.$$

Proof

For each θ, the cut I_R^θ is determined by the subset

$$\{v \in \operatorname{Im}(\rho_R) : \theta \preceq_R v\}.$$

Since $(Z_R, \preceq_R)$ is a chain, these subsets are nested upper segments of $\operatorname{Im}(\rho_R)$.

A finite chain with m image values has at most $m + 1$ upper segments.

Therefore there are at most $|\operatorname{Im}(\rho_R)| + 1$ distinct cuts.

The organization lattice at θ is determined by the cut I_R^θ, so the same bound holds for distinct organization lattices.

$\square$

Definition 13.4: Threshold Support of an Organization

For

$$O = (S, T) \in \mathcal{P}(A_R) \times \mathcal{P}(Q_R),$$

define

$$\Theta_R(O) := \{\theta \in Z_R^\circ : O \in \mathrm{Org}_\theta^\circ(R)\}.$$

Definition 13.5: Persistence Diagram

Assume $A_R \times Q_R$ is finite, $(Z_R, \preceq_R)$ is a chain, and $\mathrm{Im}(\rho_R)$ is finite.

The persistence diagram of R is

$$\mathrm{Pers}(R) := \{(O, \Theta_R(O)) : O \in \bigcup_{\theta \in Z_R^\circ} \mathrm{Org}_\theta^\circ(R),\ \Theta_R(O) \neq \emptyset\}.$$

Theorem 13.6: Persistence Well-Definedness

Under the hypotheses of Definition 13.5,

$$\mathrm{Pers}(R)$$

is finite.

Proof

Since $A_R \times Q_R$ is finite, both A_R and Q_R are finite.

Therefore

$$\mathcal{P}(A_R) \times \mathcal{P}(Q_R)$$

is finite.

Thus the set of possible pairs $O = (S, T)$ is finite.

Therefore $\mathrm{Pers}(R)$ is finite.

$$\square$$

Definition 13.7: Finite-Chain Persistence Length

Let $(Z_R, \preceq_R)$ be a finite chain.

For $X \subseteq Z_R$, define

$$\lambda_R(X) := |X|.$$

For

$$O \in \bigcup_{\theta \in Z_R^\circ} \mathrm{Org}_\theta^\circ(R),$$

define

$$\mathrm{plen}_R(O) := \lambda_R(\Theta_R(O)).$$

Definition 13.8: Real-Interval Persistence Length

If $Z_R \subseteq \mathbb{R}$, define

$$\mathrm{plen}_R(O)$$

as the Lebesgue measure of $\Theta_R(O)$, when $\Theta_R(O)$ is measurable.

---

14. Persistence-Support Formula

Assume in this section that A_R, Q_R are finite nonempty sets and $(Z_R, \preceq_R)$ is a chain.

Let

$$O = (S, T)$$

with

$$\emptyset \neq S \subsetneq A_R, \quad \emptyset \neq T \subsetneq Q_R.$$

Definition 14.1: Internal Support Height

$$U_R(S, T) := \bigwedge\{\rho_R(a, q) : a \in S, q \in T\}.$$

Definition 14.2: External Answer Obstruction

$$L_A(S, T) := \bigvee_{a \in A_R \setminus S} \bigwedge_{q \in T} \rho_R(a, q).$$

Definition 14.3: External Question Obstruction

$$L_Q(S, T) := \bigvee_{q \in Q_R \setminus T} \bigwedge_{a \in S} \rho_R(a, q).$$

Definition 14.4: External Obstruction Height

$$L_R(S, T) := L_A(S, T) \vee L_Q(S, T).$$

Theorem 14.5: Persistence-Support Interval Formula

For $O = (S, T)$ as above,

$$O \in \mathrm{Org}^{\circ}_{\theta}(R)$$

if and only if

$$L_R(S, T) \prec_R \theta \preceq_R U_R(S, T).$$

Therefore

$$\Theta_R(O) = \{\theta \in Z_R^\circ : L_R(S,T) \prec_R \theta \preceq_R U_R(S,T)\}.$$

Proof

First assume

$$O \in \mathrm{Org}_\theta^\circ(R).$$

Then

$$T = S^{\uparrow_R^\theta}.$$

For every $q \in T$ and every $a \in S$,

$$\theta \preceq_R \rho_R(a,q).$$

Thus

$$\theta \preceq_R U_R(S,T).$$

For every $q \in Q_R \setminus T$, since $q \notin S^{\uparrow_R^\theta}$, there exists $a \in S$ such that

$$\neg(\theta \preceq_R \rho_R(a,q)).$$

Because Z_R is a chain,

$$\rho_R(a,q) \prec_R \theta.$$

Thus

$$\bigwedge_{a \in S} \rho_R(a,q) \prec_R \theta.$$

Taking the join over $q \in Q_R \setminus T$ gives

$$L_Q(S,T) \prec_R \theta.$$

Similarly, since

$$S = T^{\downarrow_R^\theta},$$

for every $a \in A_R \setminus S$, there exists $q \in T$ such that

$$\rho_R(a,q) \prec_R \theta.$$

Thus

$$L_A(S,T) \prec_R \theta.$$

Therefore

$$L_R(S,T) \prec_R \theta \preceq_R U_R(S,T).$$

Conversely, assume

$$L_R(S,T) \prec_R \theta \preceq_R U_R(S,T).$$

Since $\theta \preceq_R U_R(S,T)$, for every $a \in S$ and $q \in T$,

$$\theta \preceq_R \rho_R(a,q).$$

Thus

$$T \subseteq S^{\uparrow_R^\theta}$$

and

$$S \subseteq T^{\downarrow_R^\theta}.$$

If $q \in Q_R \setminus T$, then

$$\bigwedge_{a \in S} \rho_R(a,q) \preceq_R L_Q(S,T) \prec_R \theta.$$

Since the set S is finite, there exists $a \in S$ such that

$$\rho_R(a,q) \prec_R \theta.$$

Thus

$$q \notin S^{\uparrow_R^\theta}.$$

So

$$S^{\uparrow_R^\theta} = T.$$

Similarly, if $a \in A_R \setminus S$, then

$$\bigwedge_{q \in T} \rho_R(a,q) \preceq_R L_A(S,T) \prec_R \theta.$$

Since the set T is finite, there exists $q \in T$ such that

$$\rho_R(a,q) \prec_R \theta.$$

Thus

$$a \notin T^{\downarrow_R^\theta}.$$

So

$$T^{\downarrow_R^\theta} = S.$$

Thus

$$O \in \mathrm{Org}_\theta(R).$$

Since S and T are both nonempty proper subsets, O is not $\bot_\theta(R)$ or $\top_\theta(R)$.
Therefore

$$O \in \mathrm{Org}_\theta^\circ(R).$$

$\square$

Corollary 14.6: Real-Valued Persistence Length

Assume the hypotheses of Theorem 14.5 and additionally

$$Z_R \subseteq \mathbb{R}$$

with the usual order.
Then

$$\mathrm{plen}_R(S,T) = \max\{0, U_R(S,T) - L_R(S,T)\}$$

when Z_R is a real interval and Z_R° contains $(L_R(S,T), U_R(S,T)]$.

Proof

By Theorem 14.5,

$$\Theta_R(S,T) = (L_R(S,T), U_R(S,T)] \cap Z_R^\circ.$$

The Lebesgue measure of $(L_R(S,T), U_R(S,T)]$ is

$$U_R(S,T) - L_R(S,T)$$

when $L_R(S,T) < U_R(S,T)$, and zero otherwise.

$\square$

Definition 14.7: Candidate Pair Family

Assume A_R, Q_R are finite nonempty sets.

Define

$$\mathcal{C}_R := \{(S, T) : \emptyset \neq S \subsetneq A_R,\ \emptyset \neq T \subsetneq Q_R\}.$$

Definition 14.8: Real-Valued Organization Bar

Assume

$$Z_R = [0, 1]$$

with the usual order.

For $O = (S, T) \in \mathcal{C}_R$, define the organization bar

$$B_R(O) := (L_R(O), U_R(O)]$$

when

$$L_R(O) < U_R(O),$$

and define

$$B_R(O) := \emptyset$$

when

$$L_R(O) \geq U_R(O).$$

Definition 14.9: Labeled Barcode

Assume $Z_R = [0, 1]$ and A_R, Q_R are finite nonempty sets.

The labeled barcode of R is

$$\mathrm{Bar}^{\mathrm{lab}}(R) := \{(O, B_R(O)) : O \in \mathcal{C}_R\}.$$

Definition 14.10: Unlabeled Barcode

Assume $Z_R = [0, 1]$ and A_R, Q_R are finite nonempty sets.

The unlabeled barcode of R is the finite multiset

$$\mathrm{Bar}(R) := \{B_R(O) : O \in \mathcal{C}_R,\ B_R(O) \neq \emptyset\}.$$

Multiplicity is inherited from the number of candidate pairs yielding the same nonempty interval.

Theorem 14.11: Barcode Reconstruction of Proper Organizations

Assume $Z_R = [0,1]$ and A_R, Q_R are finite nonempty sets.

For every $\theta \in (0,1)$,

$$\mathrm{Org}^{\circ}_{\theta}(R) = \{O \in \mathcal{C}_R : \theta \in B_R(O)\}.$$

Proof

Let $O \in \mathcal{C}_R$.

By Theorem 14.5,

$$O \in \mathrm{Org}^{\circ}_{\theta}(R) \quad \Longleftrightarrow \quad L_R(O) < \theta \le U_R(O).$$

By Definition 14.8, this is equivalent to

$$\theta \in B_R(O).$$

Since Theorem 14.5 applies to all $O \in \mathcal{C}_R$, the equality follows.

$\square$

---

15. Evaluated Organization Structures

Definition 15.1: Evaluation Poset

An evaluation poset is a partially ordered set

$$(W, \le_W).$$

Definition 15.2: Evaluator

For $\theta \in Z_R$, an evaluator is a map

$$E_R^{\theta} : \mathrm{Org}^{\circ}_{\theta}(R) \to W.$$

Definition 15.3: Evaluated Organization Structure

An evaluated organization structure is a pair

$$(R, E)$$

where R is an organization structure and

$$E = \{E_R^{\theta} : \theta \in Z_R^{\circ}\}$$

is a family of evaluators with common evaluation poset $(W, \le_W)$.

Definition 15.4: Persistence Evaluator

Assume $\mathrm{plen}_R(O)$ is defined for each proper organization appearing at some threshold.

The persistence evaluator is

$$E^{\theta}_{\mathrm{pers},R} : \mathrm{Org}^{\circ}_{\theta}(R) \to [0, \infty]$$

given by

$$E^{\theta}_{\mathrm{pers},R}(O) := \mathrm{plen}_R(O).$$

Definition 15.5: Support-Size Evaluator

For

$$O = (S, T) \in \mathrm{Org}^{\circ}_{\theta}(R),$$

define

$$E^{\theta}_{\mathrm{sup},R}(S, T) := |S| \cdot |T|$$

when A_R, Q_R are finite.

Definition 15.6: Persistence-Support Evaluator

When both evaluators are defined, set

$$E^{\theta}_{\mathrm{ps},R}(S, T) := (\mathrm{plen}_R(S, T), |S| \cdot |T|)$$

with lexicographic order on

$$[0, \infty] \times \mathbb{N}.$$

Definition 15.7: Evaluator Maximizer

If $(W, \leq_W)$ is a total order, then

$$O \in \mathrm{Org}^{\circ}_{\theta}(R)$$

is an evaluator maximizer when

$$\forall O' \in \mathrm{Org}^{\circ}_{\theta}(R), \quad E^{\theta}_{R}(O') \leq_W E^{\theta}_{R}(O).$$

Theorem 15.8: Finite Maximizer Existence

If

$$\mathrm{Org}^{\circ}_{\theta}(R)$$

is finite and nonempty, and $(W, \leq_W)$ is a total order, then an evaluator maximizer exists.

Proof

The image

$$E_R^{\theta}(\mathrm{Org}^{\circ}_{\theta}(R))$$

is a finite nonempty subset of a total order.

It has a maximal element.

Any organization mapped to that maximal element is a maximizer.

$$\square$$

Theorem 15.9: Persistence Robustness Under Threshold Perturbation

Assume $(Z_R, \preceq_R)$ is a chain, and let

$$O = (S, T)$$

satisfy the hypotheses of Theorem 14.5.

If

$$O \in \mathrm{Org}^{\circ}_{\theta}(R)$$

and

$$L_R(S,T) \prec_R \phi \preceq_R U_R(S,T),$$

then

$$O \in \mathrm{Org}^{\circ}_{\phi}(R).$$

Proof

By Theorem 14.5,

$$O \in \mathrm{Org}^{\circ}_{\eta}(R) \quad \Longleftrightarrow \quad L_R(S,T) \prec_R \eta \preceq_R U_R(S,T).$$

Apply this with $\eta = \phi$.

$$\square$$

Corollary 15.10: Approximation-Enough Criterion for Persistence Evaluators

Assume $(Z_R, \preceq_R)$ is a chain, and let $O = (S,T)$ satisfy the hypotheses of Theorem 14.5.

If a threshold estimate $\hat{\theta}$ satisfies

$$L_R(S,T) \prec_R \hat{\theta} \preceq_R U_R(S,T),$$

then replacing θ by $\hat{\theta}$ does not change whether O is a proper organization.

If, additionally, O is a persistence-evaluator maximizer at every

$$\eta \in \Theta_R(O),$$

then replacing θ by $\hat{\theta}$ does not change the persistence-evaluator-selected organization.

Proof

The first statement is Theorem 15.9.

The second statement follows directly from the definition of evaluator maximizer applied at each $\eta \in \Theta_R(O)$.

□

Definition 15.11: Support-Monotone Persistence

Assume real-valued persistence length and finite faces.

The structure R has support-monotone persistence when for all proper organizations $O = (S,T)$ and $O' = (S',T')$,

$$|S| \cdot |T| < |S'| \cdot |T'| \quad \Longrightarrow \quad \mathrm{plen}_R(O) < \mathrm{plen}_R(O').$$

Theorem 15.12: Agreement of Persistence and Support Maximizers Under Support-Monotone Persistence

Assume R has support-monotone persistence.

Let $O = (S,T) \in \mathrm{Org}^{\circ}_{\theta}(R)$ maximize the support-size evaluator.

Assume additionally that for every $O' = (S',T') \in \mathrm{Org}^{\circ}_{\theta}(R)$,

$$E^{\theta}_{\mathrm{sup},R}(O') = E^{\theta}_{\mathrm{sup},R}(O) \quad \Longrightarrow \quad E^{\theta}_{\mathrm{pers},R}(O') \leq E^{\theta}_{\mathrm{pers},R}(O).$$

Then O also maximizes the persistence evaluator.

Proof

Let $O' = (S',T') \in \mathrm{Org}^{\circ}_{\theta}(R)$.

Since O maximizes support size,

$$|S'| \cdot |T'| \leq |S| \cdot |T|.$$

If

$$|S'| \cdot |T'| < |S| \cdot |T|,$$

then support-monotone persistence gives

$$\mathrm{plen}_R(O') < \mathrm{plen}_R(O).$$

If

$$|S'| \cdot |T'| = |S| \cdot |T|,$$

then the equality-tie hypothesis gives

$$\mathrm{plen}_R(O') \leq \mathrm{plen}_R(O).$$

Therefore every proper organization at θ has persistence no greater than O.

Thus O maximizes the persistence evaluator.

$\square$

Corollary 15.13: Strict Support-Functional Persistence Agreement

If there exists a strictly increasing map

$$h : \mathbb{N} \to [0, \infty]$$

such that for every proper organization $O = (S, T)$,

$$\mathrm{plen}_R(O) = h(|S| \cdot |T|),$$

then the support-size and persistence maximizers coincide.

Proof

For any two proper organizations O, O',

$$|S| \cdot |T| \leq |S'| \cdot |T'| \iff h(|S| \cdot |T|) \leq h(|S'| \cdot |T'|).$$

Thus both evaluators induce the same ordering.

$\square$

Definition 15.14: Support-Only Evaluator

Assume finite faces.

An evaluator E_R^θ is support-only when there exists a map

$$\psi : \mathbb{N} \to W$$

such that for every $O = (S, T) \in \mathrm{Org}_\theta^\circ(R)$,

$$E_R^\theta(O) = \psi(|S| \cdot |T|).$$

A support-only evaluator is support-monotone when

$$n \leq m \Rightarrow \psi(n) \leq_W \psi(m).$$

Theorem 15.15: Support-Only Non-Uniqueness No-Go

Assume finite faces and a total evaluation order.

If there exist distinct

$$O_1 = (S_1, T_1), \qquad O_2 = (S_2, T_2)$$

in $\mathrm{Org}_\theta^\circ(R)$ such that

$$|S_1| \cdot |T_1| = |S_2| \cdot |T_2|$$

and this common support size is maximal among all proper organizations at θ, then no support-monotone support-only evaluator can uniquely select between O_1 and O_2.

Proof

Let E_R^θ be support-monotone and support-only.

Then there exists $\psi : \mathbb{N} \to W$ with

$$E_R^\theta(O) = \psi(|S| \cdot |T|).$$

Since O_1 and O_2 have equal support size,

$$E_R^\theta(O_1) = E_R^\theta(O_2).$$

Since their common support size is maximal and ψ is monotone, no organization with smaller support size has strictly larger evaluator value.

Thus O_1 is a maximizer if and only if O_2 is a maximizer.

Therefore the evaluator cannot uniquely select between O_1 and O_2.

$\square$

Example 15.16: Divergence of Persistence and Support Maximizers

Let

$$A_R = \{a_1, a_2, a_3\}, \quad Q_R = \{q_1, q_2, q_3\}, \quad Z_R = [0, 1].$$

Define ρ_R by the matrix

	q_1	q_2	q_3
a_1	0.90	0.10	0.10
a_2	0.10	0.75	0.75
a_3	0.10	0.75	0.75

For

$$O_1 = (\{a_1\}, \{q_1\}),$$

$$U_R(O_1) = 0.90, \quad L_R(O_1) = 0.10, \quad \mathrm{plen}_R(O_1) = 0.80, \quad E_{\mathrm{sup}}(O_1) = 1.$$

For

$$O_2 = (\{a_2, a_3\}, \{q_2, q_3\}),$$

$$U_R(O_2) = 0.75, \quad L_R(O_2) = 0.10, \quad \mathrm{plen}_R(O_2) = 0.65, \quad E_{\mathrm{sup}}(O_2) = 4.$$

Thus the persistence evaluator selects O_1, while the support-size evaluator selects O_2.

Definition 15.17: Relation-Internal Task

A relation-internal task on R is an ordered pair

$$\tau = (P_\tau, N_\tau)$$

where

$$P_\tau, N_\tau \subseteq R$$

are finite and

$$P_\tau \cap N_\tau = \emptyset.$$

The positive answer face of τ is

$$A_\tau^+ := \alpha_R[P_\tau].$$

The positive question face of τ is

$$Q_\tau^+ := \chi_R[P_\tau].$$

The negative answer face of τ is

$$A_\tau^- := \alpha_R[N_\tau].$$

The negative question face of τ is

$$Q_\tau^- := \chi_R[N_\tau].$$

Definition 15.18: Task Satisfaction

Let

$$\tau = (P_\tau, N_\tau)$$

be a relation-internal task on R.

For

$$O = (S, T) \in \mathrm{Org}^\circ_\theta(R),$$

write

$$O \vDash_R \tau$$

when

$$A_\tau^+ \subseteq S, \quad Q_\tau^+ \subseteq T,$$

and

$$A_\tau^- \cap S = \emptyset, \quad Q_\tau^- \cap T = \emptyset.$$

Define

$$\mathrm{Org}^\circ_{\theta,\tau}(R) := \{O \in \mathrm{Org}^\circ_\theta(R) : O \vDash_R \tau\}.$$

Definition 15.19: Task-Relative Evaluator

Let

$$(W, \leq_W)$$

be an evaluation poset.

A task-relative evaluator for τ at θ is a map

$$E^\theta_{R,\tau} : \mathrm{Org}^\circ_{\theta,\tau}(R) \to W.$$

Definition 15.20: Task-Relative Maximizer

If $(W, \leq_W)$ is a total order, then

$$O \in \mathrm{Org}^{\circ}_{\theta,\tau}(R)$$

is a task-relative maximizer when

$$\forall O' \in \mathrm{Org}^{\circ}_{\theta,\tau}(R), \quad E^{\theta}_{R,\tau}(O') \leq_W E^{\theta}_{R,\tau}(O).$$

Theorem 15.21: Relation-Internality of Tasks

For every relation-internal task

$$\tau = (P_\tau, N_\tau),$$

all task-satisfaction data

$$A^+_\tau, Q^+_\tau, A^-_\tau, Q^-_\tau$$

are obtained from R, α_R, and χ_R.

Proof

By Definition 15.17,

$$P_\tau, N_\tau \subseteq R.$$

By Definition 2.2,

$$\alpha_R : R \to A_R.$$

By Definition 2.3,

$$\chi_R : R \to Q_R.$$

Therefore

$$A^+_\tau = \alpha_R[P_\tau], \quad Q^+_\tau = \chi_R[P_\tau], \quad A^-_\tau = \alpha_R[N_\tau], \quad Q^-_\tau = \chi_R[N_\tau]$$

are determined by the primitive carrier and its quotient maps.

□

Theorem 15.22: Finite Task-Relative Maximizer Existence

If

$$\mathrm{Org}^{\circ}_{\theta,\tau}(R)$$

is finite and nonempty, and $(W, \leq_W)$ is a total order, then every task-relative evaluator

$$E^{\theta}_{R,\tau} : \mathrm{Org}^{\circ}_{\theta,\tau}(R) \to W$$

has a task-relative maximizer.

Proof

The image

$$E^{\theta}_{R,\tau}(\mathrm{Org}^{\circ}_{\theta,\tau}(R))$$

is a finite nonempty subset of a total order.

It has a maximal element.

Any organization mapped to that maximal element is a task-relative maximizer.

$$\square$$

Theorem 15.23: Task Stability Under Threshold Perturbation

Assume $(Z_R, \preceq_R)$ is a chain, and let

$$O = (S, T)$$

satisfy the hypotheses of Theorem 14.5.

Let τ be a relation-internal task.

If

$$O \in \mathrm{Org}^{\circ}_{\theta,\tau}(R)$$

and

$$L_R(S,T) \prec_R \phi \preceq_R U_R(S,T),$$

then

$$O \in \mathrm{Org}^{\circ}_{\phi,\tau}(R).$$

Proof

Since

$$O \in \mathrm{Org}^{\circ}_{\theta,\tau}(R),$$

it follows that

$$O \in \mathrm{Org}^{\circ}_{\theta}(R)$$

and

$$O \vDash_R \tau.$$

By Theorem 15.9 and

$$L_R(S,T) \prec_R \phi \preceq_R U_R(S,T),$$

hence

$$O \in \mathrm{Org}^{\circ}_{\phi}(R).$$

The relation

$$O \vDash_R \tau$$

depends only on $S, T, A^+_\tau, Q^+_\tau, A^-_\tau, Q^-_\tau$, not on θ.
Therefore

$$O \in \mathrm{Org}^{\circ}_{\phi,\tau}(R).$$

□

Corollary 15.24: Task-Relative Approximation-Enough Criterion

Assume the hypotheses of Theorem 15.23.
If a threshold estimate $\hat{\theta}$ satisfies

$$L_R(S,T) \prec_R \hat{\theta} \preceq_R U_R(S,T),$$

then replacing θ by $\hat{\theta}$ does not change whether O is a proper organization satisfying τ.
If, additionally, O is a task-relative maximizer at every

$$\eta \in \Theta_R(O),$$

then replacing θ by $\hat{\theta}$ does not change the task-relative selected organization.

Proof

The first statement is Theorem 15.23.

The second statement follows from Definition 15.20 at each $\eta \in \Theta_R(O)$.

$\square$

---

16. Functoriality of Persistence Under Cut-Isomorphism Propagation

Theorem 16.1: Transported Derivation Under Cut-Isomorphism

Let

$$m = (f, g, k) : R \to R'$$

be a cut-isomorphism propagation morphism.

For every $S \subseteq A_R$ and $\theta \in Z_R$,

$$(f[S])^{\uparrow^{k(\theta)}_{R'}} = g^{-1}[S^{\uparrow^{\theta}_{R}}].$$

For every $T \subseteq Q_R$ and $\theta \in Z_R$,

$$(g^{-1}[T])^{\downarrow^{k(\theta)}_{R'}} = f[T^{\downarrow^{\theta}_{R}}].$$

Proof

First prove the up-derivation identity.

Let $q' \in Q_{R'}$.

$$q' \in (f[S])^{\uparrow^{k(\theta)}_{R'}}$$

if and only if

$$\forall a' \in f[S],\ a' \Vdash^{k(\theta)}_{R'} q'.$$

Since f is bijective onto $A_{R'}$, this is equivalent to

$$\forall a \in S,\ f(a) \Vdash^{k(\theta)}_{R'} q'.$$

By Corollary 12.8,

$$f(a) \Vdash^{k(\theta)}_{R'} q' \quad \Longleftrightarrow \quad a \Vdash^{\theta}_{R} g(q').$$

Thus the condition is equivalent to

$$\forall a \in S,\ a \Vdash^{\theta}_{R} g(q'),$$

which is equivalent to

$$g(q') \in S^{\uparrow_R^\theta},$$

which is equivalent to

$$q' \in g^{-1}[S^{\uparrow_R^\theta}].$$

The down-derivation identity is dual.

$$\square$$

Theorem 16.2: Transport of Organizations

Let

$$m = (f, g, k) : R \to R'$$

be a cut-isomorphism propagation morphism.

For every $O = (S, T)$ and every $\theta \in Z_R$,

$$O \in \mathrm{Org}_\theta(R) \quad \Longleftrightarrow \quad m_*(O) \in \mathrm{Org}_{k(\theta)}(R').$$

If $O \in \mathrm{Org}^\circ_\theta(R)$, then

$$m_*(O) \in \mathrm{Org}^\circ_{k(\theta)}(R').$$

Proof

Let $O = (S, T)$.

By Definition 12.16,

$$m_*(O) = (f[S], g^{-1}[T]).$$

By Theorem 16.1,

$$(f[S])^{\uparrow_{R'}^{k(\theta)}} = g^{-1}[S^{\uparrow_R^\theta}],$$

and

$$(g^{-1}[T])^{\downarrow_{R'}^{k(\theta)}} = f[T^{\downarrow_R^\theta}].$$

Thus

$$S = T^{\downarrow_R^\theta} \quad \text{and} \quad T = S^{\uparrow_R^\theta}$$

if and only if

$$f[S] = (g^{-1}[T])^{\downarrow_{R'}^{k(\theta)}} \quad \text{and} \quad g^{-1}[T] = (f[S])^{\uparrow_{R'}^{k(\theta)}}.$$

Therefore organization is transported and reflected.

If O is proper, then $S \neq A_R$ and $T \neq Q_R$ for the boundary cases. Since f and g are bijective, transported boundary organizations are exactly boundary organizations. Hence non-boundary organizations transport to non-boundary organizations.

$\square$

Theorem 16.3: Pullback Equality of Threshold Supports

Let

$$m = (f, g, k) : R \to R'$$

be a cut-isomorphism propagation morphism.

For every $O \in \mathcal{P}(A_R) \times \mathcal{P}(Q_R)$ and every $\theta \in Z_R^\circ$,

$$\theta \in \Theta_R(O) \quad \Longleftrightarrow \quad k(\theta) \in \Theta_{R'}(m_*(O)).$$

Equivalently,

$$\Theta_R(O) = Z_R^\circ \cap k^{-1}[\Theta_{R'}(m_*(O))].$$

Proof

Since k is interior-preserving,

$$\theta \in Z_R^\circ \quad \Longrightarrow \quad k(\theta) \in Z_{R'}^\circ.$$

By Theorem 16.2,

$$O \in \mathrm{Org}_\theta^\circ(R) \quad \Longleftrightarrow \quad m_*(O) \in \mathrm{Org}_{k(\theta)}^\circ(R').$$

This is exactly the claimed threshold-support equivalence.

$\square$

Corollary 16.4: Persistence-Length Preservation Under Finite-Chain Cut-Isomorphism

Assume $Z_R, Z_{R'}$ are finite chains and $k : Z_R \to Z_{R'}$ is an interior-preserving order embedding.

If, for a given O,

$$\Theta_{R'}(m_*(O)) \subseteq k[Z_R^\circ],$$

then

$$\mathrm{plen}_R(O) = \mathrm{plen}_{R'}(m_*(O)).$$

Proof

By Theorem 16.3,

$$k[\Theta_R(O)] \subseteq \Theta_{R'}(m_*(O)).$$

For thresholds in $k[Z_R^\circ]$, Theorem 16.3 gives exact equivalence.

If $\Theta_{R'}(m_*(O)) \subseteq k[Z_R^\circ]$, then

$$\Theta_{R'}(m_*(O)) = k[\Theta_R(O)].$$

Since order embeddings are injective,

$$|\Theta_R(O)| = |k[\Theta_R(O)]|.$$

Therefore

$$|\Theta_R(O)| = |\Theta_{R'}(m_*(O))|.$$

Thus persistence length is preserved.

$\square$

Definition 16.5: Task Profile

For a relation-internal task τ, define its task profile

$$\Pi_R(\tau) := (A_\tau^+, Q_\tau^+, A_\tau^-, Q_\tau^-).$$

Definition 16.6: Transported Task Profile

Let

$$m = (f, g, k) : R \to R'$$

be a cut-isomorphism propagation morphism.

A relation-internal task τ' on R' is an m-transport of τ when

$$A_{\tau'}^+ = f[A_\tau^+], \qquad Q_{\tau'}^+ = g^{-1}[Q_\tau^+],$$

$$A_{\tau'}^- = f[A_\tau^-], \qquad Q_{\tau'}^- = g^{-1}[Q_\tau^-].$$

Theorem 16.7: Task Satisfaction Transport

Let $m : R \to R'$ be a cut-isomorphism propagation morphism.

Let τ' be an m-transport of τ.

For every organization

$$O = (S, T),$$

$$O \vDash_R \tau \quad \Longleftrightarrow \quad m_*(O) \vDash_{R'} \tau'.$$

Proof

By Definition 15.18,

$$O \vDash_R \tau$$

means

$$A_\tau^+ \subseteq S, \quad Q_\tau^+ \subseteq T,$$

and

$$A_\tau^- \cap S = \emptyset, \quad Q_\tau^- \cap T = \emptyset.$$

Since

$$m_*(O) = (f[S], g^{-1}[T]),$$

and since f and g are bijections, these four conditions are equivalent to

$$f[A_\tau^+] \subseteq f[S], \quad g^{-1}[Q_\tau^+] \subseteq g^{-1}[T],$$

and

$$f[A_\tau^-] \cap f[S] = \emptyset, \quad g^{-1}[Q_\tau^-] \cap g^{-1}[T] = \emptyset.$$

By Definition 16.6, these are exactly the conditions for

$$m_*(O) \vDash_{R'} \tau'.$$

$$\square$$

Corollary 16.8: Task-Relative Organization Transport

Under the hypotheses of Theorem 16.7, for every $\theta \in Z_R^\circ$,

$$O \in \mathrm{Org}^\circ_{\theta,\tau}(R) \quad \Longleftrightarrow \quad m_*(O) \in \mathrm{Org}^\circ_{k(\theta),\tau'}(R').$$

Proof

This is Theorem 16.2 together with Theorem 16.7.

□

Definition 16.9: Evaluator Compatibility Under Transport

Let E_R^{θ} and $E_{R'}^{k(\theta)}$ have a common evaluation poset.

They are m-compatible at θ when, for every

$$O \in \mathrm{Org}_{\theta}^{\circ}(R),$$

$$E_{R'}^{k(\theta)}(m_*(O)) = E_R^{\theta}(O).$$

Theorem 16.10: Maximizer Transport Under Compatible Evaluators

Let $m : R \to R'$ be a cut-isomorphism propagation morphism.

Assume E_R^{θ} and $E_{R'}^{k(\theta)}$ are m-compatible at θ.

If O is an evaluator maximizer for E_R^{θ}, then $m_*(O)$ is an evaluator maximizer for $E_{R'}^{k(\theta)}$.

Proof

Let

$$O' \in \mathrm{Org}_{k(\theta)}^{\circ}(R').$$

Since m is a cut-isomorphism, Theorem 16.2 gives an organization $O_0 \in \mathrm{Org}_{\theta}^{\circ}(R)$ such that

$$O' = m_*(O_0).$$

Since O maximizes E_R^{θ},

$$E_R^{\theta}(O_0) \leq_W E_R^{\theta}(O).$$

By evaluator compatibility,

$$E_{R'}^{k(\theta)}(O') = E_R^{\theta}(O_0)$$

and

$$E_{R'}^{k(\theta)}(m_*(O)) = E_R^{\theta}(O).$$

Therefore

$$E_{R'}^{k(\theta)}(O') \leq_W E_{R'}^{k(\theta)}(m_*(O)).$$

Thus $m_*(O)$ is a maximizer.

□

Corollary 16.11: Support-Size Maximizer Transport

Under cut-isomorphism propagation, support-size evaluators are compatible.

Therefore support-size maximizers transport.

Proof

Since f and g are bijections,

$$|f[S]| \cdot |g^{-1}[T]| = |S| \cdot |T|.$$

Thus support-size evaluator values are preserved.

Apply Theorem 16.10.

$\square$

Corollary 16.12: Persistence Maximizer Transport

Assume the hypotheses of Corollary 16.4 for every proper organization at θ.

Then persistence evaluators are compatible under m, and persistence maximizers transport.

Proof

Corollary 16.4 gives

$$\mathrm{plen}_R(O) = \mathrm{plen}_{R'}(m_*(O))$$

for every proper organization at θ.

Thus persistence evaluator values are preserved.

Apply Theorem 16.10.

$\square$

---

17. Stability of Persistence Intervals on Metric Zero-Boundaries

The preceding sections use order. Quantitative stability requires more than order: it requires a declared notion of zero-boundary distance.

Definition 17.1: Order-Convex Metric Chain

A metric zero-boundary chain is a structure

$$(Z, \preceq, d_Z)$$

such that $(Z, \preceq)$ is a chain and d_Z is a metric on Z.

The metric is order-convex when, for all $x, y, z \in Z$,

$$x \preceq y \preceq z$$

implies

$$d_Z(x,y) \le d_Z(x,z)$$

and

$$d_Z(y,z) \le d_Z(x,z).$$

Definition 17.2: Uniform Resolution Distance on a Metric Chain

Assume two organization structures R and R' have the same finite answer face A, the same finite question face Q, and the same metric zero-boundary chain

$$(Z, \preceq, d_Z).$$

Let

$$\rho := \rho_R, \qquad \rho' := \rho_{R'}.$$

Define

$$d_\infty(\rho, \rho') := \max_{a \in A, q \in Q} d_Z(\rho(a,q), \rho'(a,q)).$$

Definition 17.3: Labeled Endpoint Distance

For a fixed nonempty proper pair

$$O = (S,T),$$

define

$$d_I^O(R,R') := \max\{d_Z(L_R(O), L_{R'}(O)), d_Z(U_R(O), U_{R'}(O))\}.$$

Lemma 17.4: Finite Meet and Join Stability on Order-Convex Metric Chains

Let $(Z, \preceq, d_Z)$ be an order-convex metric chain. Let $x_1, \dots, x_n, y_1, \dots, y_n \in Z$, with $n \ge 1$, satisfy

$$d_Z(x_i, y_i) \le \varepsilon \quad \text{for all } i.$$

Then

$$d_Z\left(\bigwedge_i x_i, \bigwedge_i y_i\right) \le \varepsilon$$

and

$$d_Z\left(\bigvee_i x_i, \bigvee_i y_i\right) \le \varepsilon.$$

Proof

Let

$$x_* := \bigwedge_i x_i, \qquad y_* := \bigwedge_i y_i.$$

Choose j with $x_j = x_*$. If $x_* \preceq y_*$, then

$$x_* \preceq y_* \preceq y_j.$$

Order-convexity gives

$$d_Z(x_*, y_*) \le d_Z(x_*, y_j) = d_Z(x_j, y_j) \le \varepsilon.$$

If $y_* \preceq x_*$, choose k with $y_k = y_*$. Then

$$y_* \preceq x_* \preceq x_k,$$

so

$$d_Z(y_*, x_*) \le d_Z(y_*, x_k) = d_Z(y_k, x_k) \le \varepsilon.$$

Thus the meet inequality holds.

The join inequality is dual.

$\square$

Theorem 17.5: Endpoint Stability on Metric Zero-Boundaries

If

$$d_\infty(\rho, \rho') \le \varepsilon,$$

then for every nonempty proper pair $O = (S, T)$,

$$d_Z(U_R(O), U_{R'}(O)) \le \varepsilon$$

and

$$d_Z(L_R(O), L_{R'}(O)) \le \varepsilon.$$

Proof

The internal support height $U_R(O)$ is a finite meet of values $\rho(a,q)$ over $S \times T$. The corresponding $U_{R'}(O)$ is the same finite meet with ρ' in place of ρ. Lemma 17.4 gives

$$d_Z(U_R(O), U_{R'}(O)) \le \varepsilon.$$

For each fixed $a \in A \setminus S$, the term

$$\bigwedge_{q \in T} \rho(a,q)$$

changes by at most ε by Lemma 17.4. The external answer obstruction L_A is a finite join of these terms, so Lemma 17.4 again gives

$$d_Z(L_A^R(O), L_A^{R'}(O)) \le \varepsilon.$$

The same argument gives

$$d_Z(L_Q^R(O), L_Q^{R'}(O)) \le \varepsilon.$$

Since

$$L_R(O) = L_A^R(O) \vee L_Q^R(O),$$

a final application of Lemma 17.4 gives

$$d_Z(L_R(O), L_{R'}(O)) \le \varepsilon.$$

□

Corollary 17.6: Labeled Endpoint Stability

For every common finite candidate family $\mathcal{O}$, define

$$d_B^{\mathrm{lab}}(R, R'; \mathcal{O}) := \max_{O \in \mathcal{O}} d_I^O(R, R').$$

If

$$d_\infty(\rho, \rho') \le \varepsilon,$$

then

$$d_B^{\mathrm{lab}}(R, R'; \mathcal{O}) \le \varepsilon.$$

Proof

Take the maximum of Theorem 17.5 over $O \in \mathcal{O}$.

□

Definition 17.7: Metric Persistence Length

For a nonempty bar

$$B_R(O) = (L_R(O), U_R(O)]$$

in a metric chain, define

$$\operatorname{plen}_R^{d_Z}(O) := d_Z(L_R(O), U_R(O)).$$

Corollary 17.8: Metric Persistence-Length Stability

If

$$d_\infty(\rho, \rho') \le \varepsilon,$$

then, whenever $B_R(O)$ and $B_{R'}(O)$ are nonempty,

$$|\operatorname{plen}_R^{d_Z}(O) - \operatorname{plen}_{R'}^{d_Z}(O)| \le 2\varepsilon.$$

Proof

By the triangle inequality,

$$|d_Z(L_R, U_R) - d_Z(L_{R'}, U_{R'})| \le d_Z(L_R, L_{R'}) + d_Z(U_R, U_{R'}).$$

Theorem 17.5 bounds both terms by ε.

$$\square$$

Definition 17.9: Real Interval Distance

For nonempty real bars

$$I = (l, u], \qquad J = (l', u'],$$

define

$$d_I(I, J) := \max\{|l - l'|, |u - u'|\}.$$

Definition 17.10: Real Distance to the Diagonal

For a nonempty real bar

$$I = (l, u],$$

define

$$d_\Delta(I) := \frac{u - l}{2}.$$

Definition 17.11: Partial Matching of Real Barcodes

Let $\mathcal{B}$ and $\mathcal{B}'$ be finite multisets of nonempty real bars.

A partial matching μ from $\mathcal{B}$ to $\mathcal{B}'$ is a bijection between a submultiset of $\mathcal{B}$ and a submultiset of $\mathcal{B}'$.

Definition 17.12: Unlabeled Bottleneck Distance for Real Barcodes

For finite real barcodes $\mathcal{B}, \mathcal{B}'$, define

$$d_B(\mathcal{B}, \mathcal{B}')$$

as the minimum, over all partial matchings μ, of the maximum of:

$$d_I(I, \mu(I))$$

for matched bars I,

$$d_\Delta(I)$$

for unmatched bars $I \in \mathcal{B}$, and

$$d_\Delta(J)$$

for unmatched bars $J \in \mathcal{B}'$.

Theorem 17.13: Unlabeled Real Barcode Stability

Assume R and R' have the same finite answer face, the same finite question face, and

$$Z_R = Z_{R'} = [0, 1]$$

with the usual order and metric.

If

$$d_\infty(\rho, \rho') \le \varepsilon,$$

then

$$d_B(\mathrm{Bar}(R), \mathrm{Bar}(R')) \le \varepsilon.$$

Proof

Use the common candidate family

$$\mathcal{C} := \{(S, T) : \emptyset \ne S \subsetneq A,\ \emptyset \ne T \subsetneq Q\}.$$

For each $O \in \mathcal{C}$, Theorem 17.5 gives

$$|L_R(O) - L_{R'}(O)| \le \varepsilon, \qquad |U_R(O) - U_{R'}(O)| \le \varepsilon.$$

If both $B_R(O)$ and $B_{R'}(O)$ are nonempty, match them. The cost of this match is at most ε.

If $B_R(O)$ is nonempty and $B_{R'}(O) = \emptyset$, then

$$U_{R'}(O) \leq L_{R'}(O).$$

Therefore

$$U_R(O) - L_R(O) \leq 2\varepsilon.$$

Thus

$$d_\Delta(B_R(O)) \leq \varepsilon.$$

The case $B_R(O) = \emptyset$ and $B_{R'}(O) \neq \emptyset$ is identical.

The partial matching that matches bars with the same candidate label whenever both are nonempty and leaves the remaining bars unmatched has cost at most ε.

Thus the minimum matching cost is at most ε.

□

Definition 17.14: Real ε-Thickening of a Bar

For a nonempty real bar

$$I = (l, u],$$

define

$$I^\varepsilon := (l - \varepsilon, u + \varepsilon] \cap [0, 1].$$

Theorem 17.15: Real Interval-Interleaving Containment

If

$$d_\infty(\rho, \rho') \leq \varepsilon,$$

then for every candidate pair O, if both bars are nonempty, then

$$B_R(O) \subseteq B_{R'}(O)^\varepsilon$$

and

$$B_{R'}(O) \subseteq B_R(O)^\varepsilon.$$

If either bar is empty, then the nonempty bar has diagonal distance at most ε.

Proof

The endpoint inequalities of Theorem 17.5 imply that each endpoint of one nonempty bar lies within ε of the corresponding endpoint of the other nonempty bar.

Therefore each nonempty bar is contained in the ε-thickening of the other.

If one bar is empty, the proof of Theorem 17.13 shows that the nonempty bar has length at most 2ε, hence diagonal distance at most ε.

$\square$

Theorem 17.16: Metric-Free Quantitative Stability No-Go

The order on Z_R alone does not determine a nontrivial quantitative stability theory for persistence intervals.

More precisely, any numerical stability statement involving perturbation size, endpoint distance, bottleneck distance, or persistence length requires additional metric, valuation, or loss data on Z_R.

Proof

Order determines which cuts satisfy

$$\theta \preceq_R \rho_R(a, q),$$

and therefore determines the cut family.

It does not determine a numerical size for the difference between two values of Z_R. If $Z_R = [0, 1]$ is regarded only as an ordered set, every strictly increasing bijection

$$h : [0, 1] \to [0, 1]$$

preserves the order and hence reparametrizes the same cut structure by

$$\theta \mapsto h(\theta).$$

But such maps can change numerical endpoint distances and persistence lengths. Therefore those numerical quantities are not invariants of the order alone.

Thus quantitative stability cannot be an order-only consequence. It becomes meaningful only after a metric, valuation, or loss on Z_R is declared.

$\square$

18. Examples

18.1 Continuous Zero-Boundary Example

Let

$$A_R := \{a_1, a_2\}, \quad Q_R := \{q_1, q_2\}, \quad Z_R := [0, 1]$$

with the usual order.

Define

$$z_R^- := 0, \quad z_R^+ := 1, \quad Z_R^\circ = (0,1).$$

Define

$$\rho_R(a_1, q_1) = 0.9,$$

$$\rho_R(a_1, q_2) = 0.2,$$

$$\rho_R(a_2, q_1) = 0.2,$$

$$\rho_R(a_2, q_2) = 0.8.$$

For

$$\theta \in (0.9, 1],$$

$$I_R^\theta = \emptyset, \quad \mathrm{Org}_\theta^\circ(R) = \emptyset.$$

For

$$\theta \in (0.8, 0.9],$$

$$I_R^\theta = \{(a_1, q_1)\},$$

and

$$\mathrm{Org}_\theta^\circ(R) = \{(\{a_1\}, \{q_1\})\}.$$

For

$$\theta \in (0.2, 0.8],$$

$$I_R^\theta = \{(a_1, q_1), (a_2, q_2)\},$$

and

$$\mathrm{Org}_\theta^\circ(R) = \{(\{a_1\}, \{q_1\}), (\{a_2\}, \{q_2\})\}.$$

For

$$\theta \in [0, 0.2],$$

$$I_R^\theta = A_R \times Q_R, \quad \mathrm{Org}_\theta^\circ(R) = \emptyset.$$

Thus

$$\Theta_R(\{a_1\}, \{q_1\}) = (0.2, 0.9],$$

and

$$\Theta_R(\{a_2\}, \{q_2\}) = (0.2, 0.8].$$

Therefore

$$\mathrm{plen}_R(\{a_1\}, \{q_1\}) = 0.7,$$

and

$$\mathrm{plen}_R(\{a_2\}, \{q_2\}) = 0.6.$$

The persistence evaluator selects

$$(\{a_1\}, \{q_1\})$$

at every threshold where both proper organizations exist.

18.2 Larger Block Example

Let

$$A_R = \{a_1, a_2, a_3, a_4\}, \quad Q_R = \{q_1, q_2, q_3, q_4\}, \quad Z_R = [0, 1].$$

Let ρ_R be

	q_1	q_2	q_3	q_4
a_1	0.95	0.92	0.20	0.20
a_2	0.92	0.90	0.20	0.20
a_3	0.20	0.20	0.80	0.78
a_4	0.20	0.20	0.78	0.76

Define

$$O_A = (\{a_1, a_2\}, \{q_1, q_2\}), \quad O_B = (\{a_3, a_4\}, \{q_3, q_4\}).$$

Then

$$U_R(O_A) = 0.90, \quad L_R(O_A) = 0.20, \quad \Theta_R(O_A) = (0.20, 0.90],$$

and

$$U_R(O_B) = 0.76, \quad L_R(O_B) = 0.20, \quad \Theta_R(O_B) = (0.20, 0.76].$$

Thus

$$\mathrm{plen}_R(O_A) = 0.70, \quad \mathrm{plen}_R(O_B) = 0.56.$$

This example has multiple block organizations with distinct persistence intervals.

18.3 Diagonal Binary-Face Family

Let

$$A_R = Q_R = \{1, \dots, n\},$$

and

$$Z_R = \{z_-, z_0, z_+\}, \quad z_- \prec_R z_0 \prec_R z_+.$$

Define

$$\rho_R(i, j) = z_0 \quad \Longleftrightarrow \quad i = j,$$

and

$$\rho_R(i, j) = z_- \quad \Longleftrightarrow \quad i \neq j.$$

At $\theta = z_0$,

$$(i, j) \in I_R^{z_0} \quad \Longleftrightarrow \quad i = j.$$

The proper organizations include

$$(\{i\}, \{i\})$$

for every $1 \leq i \leq n$.

This example realizes the proof of Theorem 11.8.

---

19. Notation Index

R : primitive set of resolution occurrences.

$\equiv_A$: answer-face equivalence relation on R.

$\equiv_Q$: question-face equivalence relation on R.

$\equiv_Z$: zero-boundary-face equivalence relation on R.

$\preceq_R$: complete-lattice order on Z_R.

$$A_R := R/{\equiv_A}.$$

$$Q_R := R/{\equiv_Q}.$$

$$Z_R := R/{\equiv_Z}.$$

$$\alpha_R(r) := [r]_{\equiv_A}.$$

$$\chi_R(r) := [r]_{\equiv_Q}.$$

$$\zeta_R(r) := [r]_{\equiv_Z}.$$

$$\rho_R : A_R \times Q_R \to Z_R.$$

$$z_R^- := \bigwedge Z_R.$$

$$z_R^+ := \bigvee Z_R.$$

$$Z_R^\circ := \{z \in Z_R : z_R^- \prec_R z \prec_R z_R^+\}.$$

$$\theta : \text{zero-boundary cut}, \quad \theta \in Z_R.$$

$$a \Vdash_R^\theta q \Longleftrightarrow \theta \preceq_R \rho_R(a,q).$$

$$I_R^\theta := \{(a,q) \in A_R \times Q_R : a \Vdash_R^\theta q\}.$$

$$S^{\uparrow_R^\theta} := \{q \in Q_R : \forall a \in S,\ a \Vdash_R^\theta q\}.$$

$$T^{\downarrow_R^\theta} := \{a \in A_R : \forall q \in T,\ a \Vdash_R^\theta q\}.$$

$$C_{A,R}^\theta(S) := S^{\uparrow_R^\theta \downarrow_R^\theta}.$$

$$C_{Q,R}^\theta(T) := T^{\downarrow_R^\theta \uparrow_R^\theta}.$$

$$\mathrm{Org}_\theta(R) := \{(S,T) : S = T^{\downarrow_R^\theta} \wedge T = S^{\uparrow_R^\theta}\}.$$

$$(S,T) \leq_\theta (S',T') \iff S \subseteq S'.$$

$$\perp_\theta(R) := (Q_R^{\downarrow_R^\theta}, Q_R).$$

$$\top_\theta(R) := (A_R, A_R^{\uparrow_R^\theta}).$$

$$\mathrm{Org}_\theta^\circ(R) := \mathrm{Org}_\theta(R) \setminus \{\perp_\theta(R), \top_\theta(R)\}.$$

$$\mathrm{Im}(\rho_R) := \{\rho_R(a,q) : a \in A_R, q \in Q_R\}.$$

$$(f,g,k) : R \to R' : \text{value-coarsening propagation morphism.}$$

$$m_*(S,T) := (f[S], g^{-1}[T]).$$

$$\Theta_R(O) := \{\theta \in Z_R^\circ : O \in \mathrm{Org}_\theta^\circ(R)\}.$$

$$\mathrm{Pers}(R) := \{(O, \Theta_R(O)) : \Theta_R(O) \neq \emptyset\}.$$

$$\mathrm{plen}_R(O) : \text{persistence length of } O.$$

$$E_R^\theta : \mathrm{Org}_\theta^\circ(R) \to W : \text{evaluator.}$$

$$U_R(S,T) : \text{internal support height.}$$

$$L_R(S,T) : \text{external obstruction height.}$$

$$d_B^{\mathrm{lab}} : \text{labeled bottleneck distance.}$$

$$B_R(O) : \text{organization bar for candidate } O.$$

$$\mathrm{Bar}^{\mathrm{lab}}(R) : \text{labeled barcode.}$$

$$\mathrm{Bar}(R) : \text{unlabeled barcode.}$$

$$d_B : \text{unlabeled bottleneck distance.}$$

$$\Pi_R(\tau) : \text{task profile.}$$

---

20. Reference Anchors

Wille, R. (1982). *Restructuring lattice theory: an approach based on hierarchies of concepts.*

Ganter, B. and Wille, R. (1999). *Formal Concept Analysis: Mathematical Foundations.*

Lehmann, F. and Wille, R. (1995). *A triadic approach to formal concept analysis.*

Bělohlávek, R. (2002). *Fuzzy Relational Systems: Foundations and Principles.*

Pratt, V. R. Chu spaces and Chu transforms.

Barr, M. *-autonomous categories and Chu-space constructions.

Carlsson, G. Topology and data.

Edelsbrunner, H. and Harer, J. Computational topology and persistence.

Ghrist, R. Barcodes and applied algebraic topology.

AI-assistance disclosure

This manuscript was developed with AI-assisted drafting and review support. The author directed the research programme, selected and revised the mathematical and methodological content, and is responsible for all claims and errors.